

VeriGraph: Inspectable Claim Verification from Evidence Spans to Verdict Rules

Anonymous demonstration submission · content/layout draft

Abstract

Fact-verification systems often return a label while hiding the intermediate decisions that made it possible. **VeriGraph** turns a complex claim into typed verification obligations, retrieves source-grounded evidence, classifies sentence-level support and contradiction, and assembles an argument graph with a deterministic four-way verdict. Visitors can move from a verdict to its rule, an obligation, a model relation, and the exact source passage. The system combines local language-model decomposition, hybrid retrieval, conservative natural-language inference, linguistic auditing, and symbolic aggregation. Its contribution is an inspectable bridge between statistical components and rule-governed decisions, not another opaque factuality score.

1. Motivation and significance

A single fact-checking label compresses several different questions: Was the claim decomposed faithfully? Did retrieval miss a decisive passage? Did an NLI model over-read contradiction? Did the final rule respect conjunction, alternatives, and mixed evidence? These questions matter even when the final label is correct, yet they are difficult to ask when intermediate results are discarded or shown as unconnected tables.

VeriGraph makes the chain itself the interface. Every obligation retains exact grounding in the original claim. Every evidence candidate retains document and sentence offsets. Support and attack edges come from stored NLI relations. The final SUPPORTED, REFUTED, NOT_ENOUGH_EVIDENCE, or CONFLICTING_EVIDENCE decision follows a deterministic, composition-aware rule shown with its trace. Dataset reference labels are metadata and never enter inference.

2. System

The user selects a curated AVeriTeC case or enters a claim with source documents. A local Qwen2.5 model proposes a schema-constrained, contextualized decomposition of at most twelve standalone obligations. Each obligation has a verification role and exact UTF-16 source grounding. Invalid output receives one repair attempt; failure preserves the full claim as one obligation so the pipeline remains usable.

Retrieval segments documents and combines BGE dense similarity with lexical anchors using reciprocal-rank fusion. Exact numbers and names can recover passages that dense similarity ranks poorly. Soft source diversity avoids forcing irrelevant passages into multi-document results.

A DeBERTa model classifies each obligation/passage pair. The system defaults to neutral when a non-neutral prediction is weak. Contradiction additionally requires an inspectable anchor such as negation, incompatible numbers, opposing terms, or a matched entity-set mismatch. These gates trade recall for defensibility.

The frontend builds a typed argument graph from decomposition and NLI outputs. A backend rule engine aggregates obligation states using SINGLE, AND, or OR composition. A spaCy sidecar audits entities, qualifiers, negation, modality, and role preservation. Its warnings are deliberately verdict-neutral.

3. Interaction design

Human-titled cases can be searched and filtered by topic, verification challenge, and reference verdict. A visitor selects numerical conflict, cross-source disagreement, or insufficient evidence; opens one obligation; follows its highlighted passage into the full source; and traces that same obligation through the graph to the verdict rule. The visitor is invited to challenge a specific step rather than accept or reject the system wholesale.

4. Demonstration experience

The first minute shows the whole chain on one featured case. The presenter then asks the visitor to select a topic or challenge. Each of the five plain-language stages exposes its output and independent failure state. A three-minute path reaches the exact source and verdict trace; a five-minute research path adds decomposition auditing and a comparison between semantic retrieval and exact lexical anchors.

Live inference runs locally to avoid conference connectivity. The backup is a static interactive walkthrough containing the same typed intermediates from recorded local runs. The current snapshot has 22 cases across four reference labels. Preparation and validation target a balanced 32-case bundle, but that expansion will be claimed only after all 32 runs are recorded and audited.

5. Evaluation and limitations

Evaluation separates verdict agreement from decomposition quality and error source. Deterministic diagnostics cover atomicity, coverage, sufficiency, non-fabrication, non-redundancy, readability, and linguistic warnings. They are engineering proxies, not human judgments or the FactLens evaluator. The current 22-run snapshot has 10/22 reference-verdict agreement (45.5%), so accuracy is not presented as a contribution. Before final submission we will complete error attribution, a small human decomposition audit, and the balanced 32-case rerun.

VeriGraph verifies supplied documents; it does not establish source authority or perform open-web fact checking. Retrieval rank is not truth, NLI neutral is not a case-level insufficient-evidence label, and the graph does not infer new relations. These boundaries are stated in the interface and narration.

6. Related work and contribution

WiCE introduced real-world entailment with subclaim-oriented verification in Wikipedia contexts [1]. Wanner et al. showed that factuality evaluation is sensitive to decomposition and studied coverage, coherence, and atomicity [2]. FactLens provides a newer benchmark and evaluators for fine-grained verification, emphasizing context preservation and semantic equivalence [3]. Hu et al. demonstrated that decomposition can add downstream noise as well as benefit [4].

VeriGraph operationalizes these concerns in an interactive audit surface. Its novelty is the inspectable connection from schema-grounded obligations through conservative evidence relations to a deterministic argument and verdict trace. It does not claim novelty for the component models or linguistic parser.

7. Audience relevance

The demonstration connects natural-language processing, information retrieval, explainable AI, argumentation, and human-AI interaction. Researchers can inspect failure propagation; practitioners can distinguish a missing passage from an unsupported contradiction; educators can use the graph to show why a fact-checking label is not a primitive fact. The system is most informative when it is imperfect because disagreement has a precise location.

8. Readiness

The repository includes typed API contracts, deterministic fallbacks, automated backend and frontend tests, a preflight command, recorded walkthrough assets, and an offline runbook. The final submission gate requires three consecutive cold starts, three offline walkthrough rehearsals, and a manifest/evaluation audit before any 32-case claim is used.

References

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- [2] M. Wanner, S. Ebner, Z. Jiang, M. Dredze, and B. Van Durme. 2024. *A Closer Look at Claim Decomposition*. *SEM 2024, 153–175. doi:10.18653/v1/2024.starsem-1.13.
- [3] K. Mitra, D. Zhang, S. Rahman, and E. Hruschka. 2025. *FactLens: Benchmarking Fine-Grained Fact Verification*. Findings of ACL 2025, 18085–18096. doi:10.18653/v1/2025.findings-acl.929.
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